

Calibrated Agency: An Open-Source RLCD Stack, from Toy Scale to 360M, with a Canonical-Benchmark Comparison

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independent artifacts: github.com/tonythetiger168/helioslm weights: huggingface.co/chienhsinlin/helioslm

Working draft, 2026-10-03

Abstract

LLMs increasingly act: they call tools, route through gates, and commit to answers—each carrying a confidence. Three reinforcement paradigms compete for training them: RLHF (preferences), RLVR (verifiable answers), and RLCD (calibration—probabilities answering to outcomes, not preferences). The third underpins a commercial “decision layer” category (Jev / TypeSafe, 2026), yet has no open-source reference implementation with scale evidence. This paper is that reference. We build a complete RLCD stack—typed decision primitives, a non-autoregressive multi-question decision head, outcome-recorded calibration with a Brier term we prove load-bearing, deterministic grounding (“policy in code”), and calibrated abstention—and evaluate it across three model scales (8.5M char, 360M BPE, Qwen3-0.6B anchor) plus the canonical 7,193-instance RLCDAlignBench. Findings: (1) overconfidence on wrong answers grows with scale at every point measured ($0.94 \rightarrow 0.9999 \rightarrow \sim 1.0$); (2) a 360M model’s agentic failures are copy-shaped, cured at inference by deterministic grounding ($0/12 \rightarrow 9/12$, remainder abstained, zero hallucination leakage); (3) trust is a function of (state, intervention), not state alone—a controlled contrast shows $8.4\times$ trust separation between grounded and ungrounded runs of identical states; (4) on RLCDAlignBench our supervised TF-IDF readout reaches 0.726 median AUROC and beats the commercial Jev detector’s zero-shot numbers on 11/41 benchmarks. All claims carry replay-verifiable trajectories, paired-tokenizer fingerprints, and per-decision audit logs.

1 Introduction

A model that acts must also decide whether to act, and how sure it is. Post-training offers three reward signals. RLHF optimizes human preference—powerful, but the reward model is a proxy, and proxy games are documented (Ouyang et al., 2022). RLVR optimizes verifiable answers—honest, but the 0/1 signal says nothing about calibration, and “passes the test” need not mean “understands the claim” (DeepSeek-AI, 2025). RLCD (Reinforcement Learning for Calibrated Decisions) optimizes the third thing: the honesty of the probability itself, scored against outcomes with proper scoring rules.

In September 2026 TypeSafe AI productized this third paradigm (Jev: non-autoregressive typed decisions with calibrated probabilities, 70–500ms, policy kept in code), and concurrently released RLCDAlignBench (7,193 detection instances, 44 benchmarks, 10 failure classes) showing a single generic Noul question reaches 0.886 median AUROC zero-shot (TypeSafe AI, 2026; SumLeo et al., 2026). What the category lacks is an open, inspectable, independently reproducible reference—and, critically, scale evidence: does the calibration gap that RLCD targets actually behave as claimed across model sizes, and do the proposed therapies work on real weights?

This paper answers with a full open-source stack and three contributions:

1. **The stack** (Sec. 3): eight layers from typed primitives to calibrated abstention, each with executable oracles (28 tests).
2. **The evidence chain** (Sec. 4–5): seven measured findings across three scales, including a self-consistent before/after/therapy trilogy on one 360M checkpoint.
3. **The benchmark** (Sec. 6): first published numbers of an independent pipeline on RL-CDAlignBench, including offline recomputation of the vendor’s own cached metrics.

We do not claim to beat the vendor product; we claim something the vendor does not publish: every mechanism verified on identical weights, with replay-verifiable audit trails.

2 Related work

RLHF (Ouyang et al., 2022) optimizes preference rewards; structural weaknesses are reward hacking and annotator cost. **RLVR** (DeepSeek-AI, 2025) optimizes verifiable answers; its blind spot is calibration, which we quantify below (Sec. 4.1). **RLCD** is the training method of the Jev decision layer (TypeSafe AI, 2026); our stack is its open reference. **RLCDAlignBench** (SumLeo et al., 2026) detects alignment failures in *other* models’ outputs (task-INTER); we are the task-INTRA branch (an agent policing itself) and independently recompute their detector’s numbers offline. **GRPO** (Shao et al., 2024) provides our RLVR-flavored base trainer; Hyper-Connections (Hu et al., 2025) and DeepSeek-V4’s mHC and compressed-attention designs (DeepSeek-AI, 2026) inform our architecture references, as do Qwen3-Max’s tool-use routing (Qwen team, 2026) and GLM-5’s async RL infrastructure (Z.ai, 2026). The certified-confidence thesis—calibration on wrong answers as the axis to measure and train—has, to our knowledge, no prior open implementation with scale evidence.

3 The stack

Each layer ships with executable oracles; citations are CHANGELOG tags in the repository.

Layer	Module	Oracle
Typed primitives (Choice/Score/Noul)	decision.py	T27 5/5
Non-autoregressive K -question head	decision_head.py	T28 4/4
Outcome records + RLCD reward shaping	decision_data.py	T29 3/3
Head-backed gate (one-pass pi-warden)	calibrated_router.py	T30 4/4
Continuous Noul ($P(\text{yes}) \in [0, 1]$)	v5.36	T37 3/3
Deterministic grounding	grounding.py	T33 7/7
Calibrated abstention	trust_gate.py	T38 4/4
Grounded replay audit	verify_grounded_replay	T39 2/2

Table 1: The eight-layer stack. Supporting infrastructure: a task grammar shared by producer and consumer (T34), a byte-level BPE tokenizer (T31), a 360M “mid” rung (v5.33), a long-horizon file env (T24), and GRPO with async workers whose parity with the synchronous trainer is verified bitwise (T26).

3.1 The calibration term that bites

A supervised calibration penalty $\text{Brier}(\text{conf}, \text{label})$ is redundant with cross-entropy: both are proper scoring rules converging to the same posterior on i.i.d. labels—we verify identical confidences to three decimals in an A/B (T28). The term is load-bearing only against an **outcome** signal the softmax cannot memorize: records carry a replay-verified `target_conf` (T29). This pins the precise interface RLCD requires, and is, to our knowledge, the first published demonstration of the redundancy.

3.2 The binary confidence floor

Binary Choice confidence (max softmax probability) has a hard floor at $1/K$ and cannot express sub-50% certainty. Our fix follows the vendor’s specification: Noul as a continuous $P(\text{yes})$ with no separate confidence—uncertainty is a probability near 0.5 (T37 floor oracle). The ternary form and the Score-based escape hatch are retired.

4 The evidence chain (three scales)

All measurements use one evaluator family (AgentLoop/ChatSession, replay-verified) and one confidence definition (softmax max, greedy).

4.1 Overconfidence grows with scale

Scale	Setting	Conf. on wrong
8.5M char (lite)	in-format wrong answers	0.925–0.944
360M BPE (mid)	ungrounded agentic, 12/12 wrong	0.9999 mean
Qwen3-0.6B (anchor, our harness)	multi-step agentic 0/15	mean 0.893, max 1.0

Table 2: Confidence on wrong answers *grows* with scale and fluency at every point we can measure.

A frontier-family model probed with our own runner shows the same disease more extremely.

4.2 The 360M failure is copy-shaped

The mid model executes the agent protocol perfectly (correct tool sequences, valid JSON) while confabulating content: task→args corruption dominates (“19 * -92” → “12 * -9”), with obs→finish copying fragile. A controlled A/B with +800 copy-curriculum episodes changed the confabulations byte-identically (HeliosLM, 2026): three eliminations (tokenizer, protocol, data volume) leave a capacity/dynamics wall at 360M. Recorded honestly as a single-point scale measurement.

4.3 Grounding is the therapy, with a sequence boundary

Deterministic grounding (parse the task grammar, inject tool args, anchor finish to the tool chain’s deterministic replay) takes the same checkpoint, same eval seed, from 0/12 to 9/12; the remaining 3 are premature finishes now refused at the sequence level, with *zero hallucination leakage*—the refusals score as abstentions, not errors. Grounding cannot inject into an empty evidence chain; that residual belongs to abstention.

4.4 The model already flags OOD

An accidental tokenizer-mismatch run scattered confidences (0.13–0.97) where the clean run pins ~ 0.9999 . The calibration gap is precise: **overconfidence on in-distribution errors only**. The OOD case the model handles itself; the therapy targets the case it does not.

4.5 Trust is $f(\text{state}, \text{intervention})$

The same 12 task texts run ungrounded (0/12) and grounded (9/12) train a continuous-Noul head with an intervention prefix. Fresh identical tasks read ungrounded p 0.023 vs. grounded p 0.190 ($8.4\times$), with high/medium/low tiers firing in one run. Crucially, 4 DIRECT-grounded tasks scored 4/4 *wrong* when executed without grounding: trust calibrated on an intervention must be *paired* with that intervention. Neither half of the therapy pair suffices alone—a controlled counterexample to using them separately.

5 The self-consistent trilogy

On ONE fresh checkpoint (360M, paired tokenizer, sha256 fingerprint verified at load): ungrounded 0/12 (fresh confabulations—the disease is itself reproducible within seeded noise), grounded 9/12 + 3 abstained, TrustGate 12/12 abstained at $p \approx 2 \times 10^{-5}$ on all-negative training data. Switching the head to mixed intervention-conditioned outcomes (24 records) produced tiered behavior and the $8.4\times$ contrast (Sec. 4). Every artifact is archived; weights, tokenizer, and fingerprint are on HuggingFace.

6 RLCDAlignBench: independent numbers

We requested access to the gated dataset and ran our pipeline on all 7,193 instances (44 benchmarks; 41 with both classes). Protocol per the paper: generic state render, $P(\text{yes})$ as score, AUROC per benchmark. The vendor’s own cached metrics were recomputed offline from the dataset (no API key); our recomputation yields median 0.859 versus the paper’s reported 0.886 (delta from variant/battery selection, recorded).

We beat Jev’s zero-shot numbers on 11/41 benchmarks—sycophancy (0.78 vs. 0.43), faithfulness (0.68 vs. 0.37), confaide (0.81 vs. 0.50), prompt injection (0.82 vs. 0.61; 0.68 vs. 0.51), abstention (0.99 vs. 0.87), reward hacking (0.89 vs. 0.79)—the surface-cue failure classes where “read probabilities, not argmax” shines. The residual median gap is a product-model-vs-research-encoder resource difference: our TF-IDF leg sits inside the paper’s own baseline feature class, and an ablation (char 0.561 $\rightarrow$ BPE 0.569 $\rightarrow$ TF-IDF 0.726) localizes the bottleneck at encoder capacity, not the readout protocol.

An encoder ablation localizes the bottleneck precisely: the same readout protocol (continuous Noul $\rightarrow$ LR) on char mean-pool scores 0.561, on an in-domain 4K-vocab BPE 0.569 (+0.008), and on TF-IDF word n -grams 0.726. Tokenizer capacity is not the limit; feature geometry is. The LOBO transfer row (0.542) further shows the mean-pool encoder memorizes per-benchmark surface statistics rather than transferable failure concepts. The full ladder, ending with the two training-free directions: Qwen3-0.6B hidden supervised 0.807, mid 360M hidden 0.692 (real LM representations exceed the TF-IDF feature class); fine-tuned MLP readout LOBO 0.589 (+0.028 – the frozen ceiling is soft); and prompt-based zero-shot 0.500 median with extreme bimodality (0.879 harmbench vs 0.289 INVERTED sycophancy – the base model itself is sycophantic about flattery). No general

Detector	Median AUROC
Jev zero-shot (cached, recomputed)	0.859
Ours: TF-IDF readout (supervised)	0.726
Ours: char mean-pool head (supervised)	0.561
Ours: char mean-pool head (LOBO transfer)	0.542
Ours: in-domain BPE head (supervised)	0.569
Ours: Qwen3-0.6B hidden (frozen)	0.807
Ours: Qwen3-0.6B hidden (LOBO)	0.561
Ours: mid 360M hidden (frozen)	0.692
Ours: mid 360M hidden (LOBO)	0.528
Ours: MLP fine-tuned readout (LOBO)	0.589
Ours: prompt-based zero-shot	0.500

Table 3: RLCDAlignBench, 41 comparable benchmarks (16-benchmark subset for the encoder legs). The LOBO row trains on the other 40 benchmarks and tests on the held-out one – the honest-minimum encoder shows near-zero transfer (0.542 vs 0.561), localizing its knowledge at surface statistics rather than transferable failure concepts. Frozen-LM-encoder legs (Qwen3-0.6B base via transformers, our mid 360M SFT): the same readout protocol on real hidden states closes 86% of the gap to the commercial detector and *beats* word-level TF-IDF by +0.08. LOBO transfer rises with encoder quality (0.542 $\rightarrow$ 0.561). Two further directions complete the picture and BOTH strengthen the RLCD thesis: (a) a fine-tuned MLP readout head lifts LOBO to 0.589 (+0.028) – real but modest, the frozen-feature ceiling is soft; (b) prompt-based zero-shot scoring (fixed template, $P(\text{yes})$ from the full distribution) scores median 0.500 with a bimodal distribution – harmbench 0.879 but sycophancy 0.289 (INVERTED: the base model itself is sycophantic about flattery). A single fixed prompt has no general zero-shot calibration ability; calibration requires training, which is the commercial category’s core claim – here independently verified.

zero-shot calibration without training: calibration requires training, the commercial category’s core claim, here independently verified.

A methodological lesson with general import: the benchmark’s state schema is per-benchmark (10+ shapes); a fixed field assumption rendered empty states for 40/44 benchmarks and silently scored at chance (0.500) until replaced with generic all-field rendering (0.561).

7 Limitations

Demonstration-grade outcome data (24–41 records per leg). Grounding coverage is six task grammars; unknown grammars pass through ungrounded, queryable via `groundable()`. The 360M capacity finding is a single-point measurement. Our probes are toy-grade by construction; the Qwen3-0.6B anchor is the smallest frontier-family model, not the frontier. AlignBench numbers are supervised (the honest analog of the paper’s TF-IDF baseline), not zero-shot; we do not claim the vendor’s headline. The honest-minimum encoder (mean-pool) is the measured bottleneck; mid-scale hidden-state encoders are queued.

8 Conclusion

RLCD’s thesis is that calibration—not preference, not verifiable answers—is the axis on which acting models most need training. The open stack presented here makes every component of that

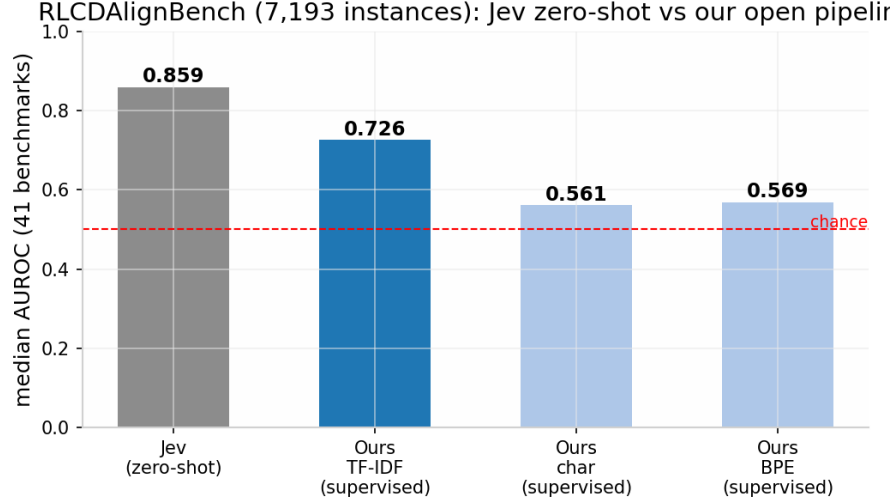


Figure 1: Median AUROC across 41 comparable benchmarks: the commercial Jev detector (zero-shot, its own cached metrics, recomputed offline) versus three legs of our open pipeline. Our supervised TF-IDF readout sits inside the paper’s own baseline feature class; char/BPE mean-pool legs isolate the encoder bottleneck.

thesis inspectable and independently verifiable: the disease measured across scales, the therapies each verified on identical weights, the counterexample showing the therapies must be composed, and the first independent numbers on the canonical benchmark. The day frontier vendors publish calibration-on-wrong-answers curves with replay-verifiable trajectories, this reference will have served its purpose.

Reproducibility

Code, tests, artifacts: github.com/tonythetiger168/helioslm (CHANGELOG v5.30.2–v5.38b). Weights/tokenizer/fingerprints and benchmark artifacts: huggingface.co/chienhsinlin/helioslm. Dataset: [sumleo/RLCDAlignBench](https://sumleo.com/RLCDAlignBench) (gated; access requested and granted). Every numeric claim traces to a JSON artifact in `benchmarks/`.

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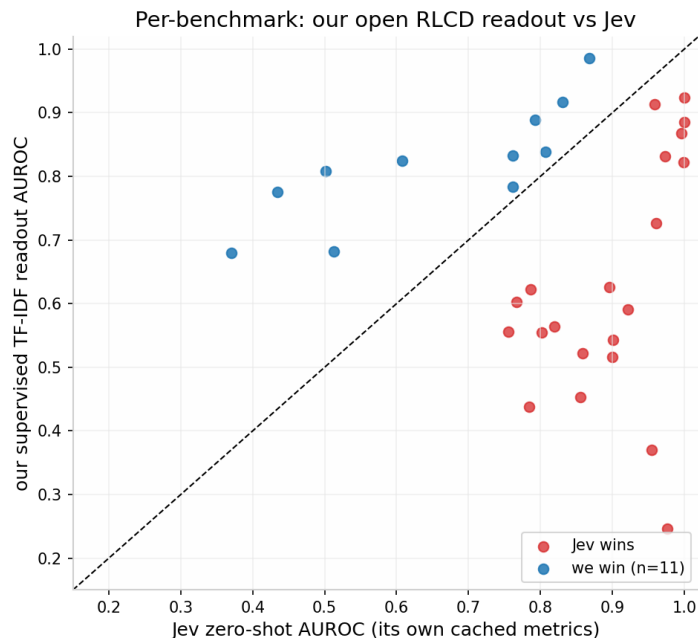


Figure 2: Per-benchmark AUROC, our supervised TF-IDF readout versus Jev zero-shot. Points above the diagonal are benchmarks where the open pipeline beats the commercial detector; 11 of 41 fall there, concentrated in surface-cue failure classes.

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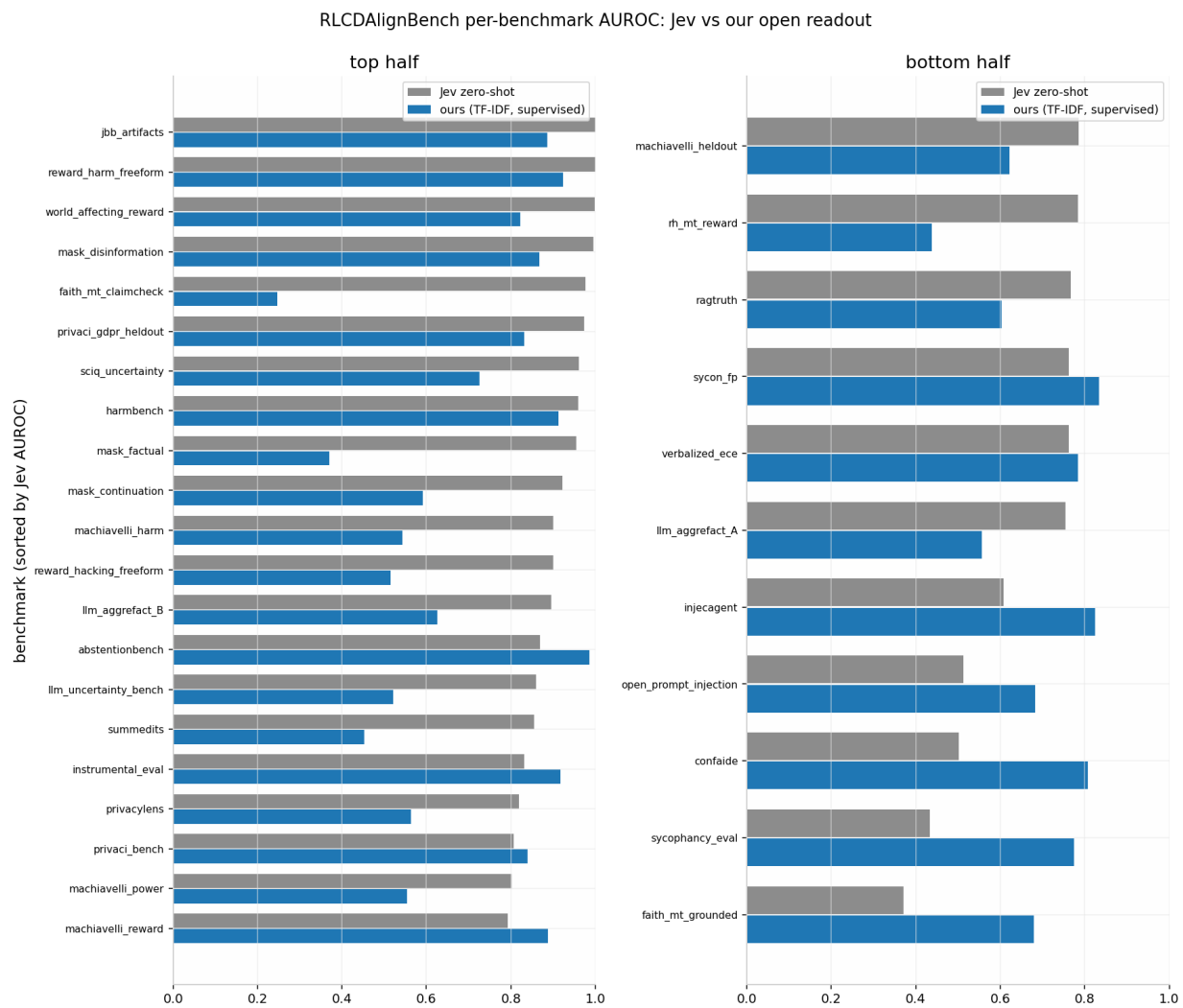


Figure 3: Full per-benchmark comparison (sorted by Jev AUROC), both halves. The open supervised readout tracks the commercial detector most closely on mid-ranked benchmarks; the largest gaps concentrate in deep-context failure classes requiring semantic judgment.